

Binary Cross Entropy (BCE) loss:

$$L = -[y \log(a) + (1-y) \log(1-a)]$$

$$\begin{aligned} \frac{dL}{da} &= - \left[\frac{d}{da}(y \log(a)) + \frac{d}{da}((1-y) \log(1-a)) \right] \\ &= - \left[y \cdot \frac{1}{a} + (1-y) \frac{d}{da}(\log(1-a)) \right] \quad [\text{Since } y \text{ is a constant}] \\ &= - \left[\frac{y}{a} + (1-y) \left(\frac{1}{1-a} \right) (-1) \right] \end{aligned}$$

$$= - \left[\frac{y}{a} - \frac{1-y}{1-a} \right]$$

$$= - \frac{y}{a} + \frac{1-y}{1-a}$$

$$= \frac{-y(1-a) + a(1-y)}{a(1-a)}$$

$$= \frac{-y + ay + a - ay}{a(1-a)}$$

$$= \frac{a-y}{a(1-a)}$$

$$\therefore \frac{dL}{da} = \frac{a-y}{a(1-a)}$$